



OR 6205: Deterministic Operations Research
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Deterministic Operations Research

Class 15: Duality Theory

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Example

From start to finish Matrix Simplex Operation

$$\text{Max } z = 4x_1 + 3x_2 + 6x_3$$

S.T.

$$3x_1 + 1x_2 + 3x_3 \leq 30$$

$$2x_1 + 2x_2 + 3x_3 \leq 40$$

$$x_1 \geq 0; x_2 \geq 0; x_3 \geq 0$$

I want to know if you understand the idea that:

1. We are finding at each iteration one of the extreme points of the feasible region.
2. We find the point by solving a set of n equations at a time.
3. One way to do it is by doing the Elementary Row Operations (ERO), other way is by finding the Inverse of the basis formed by the basic variables.
4. To move from one point to the other and make sure it is still feasible (in the equality constraints/variables we are dropping), we set the variables associated with those constraints to zero. The basic variable are the variables related to each constraint, or the Non-basic variables are the ones that are zero at each constraint.
5. We know we reach the optimal when dropping one constraint/variable to enter one other does not improve my Z value.
6. We can guarantee it because both feasible and objective function are convex sets.

Graphical Feasible Region

Matrix LP form

$$\text{Max } z = cx$$

$$\text{s.t. } Ax \leq b$$

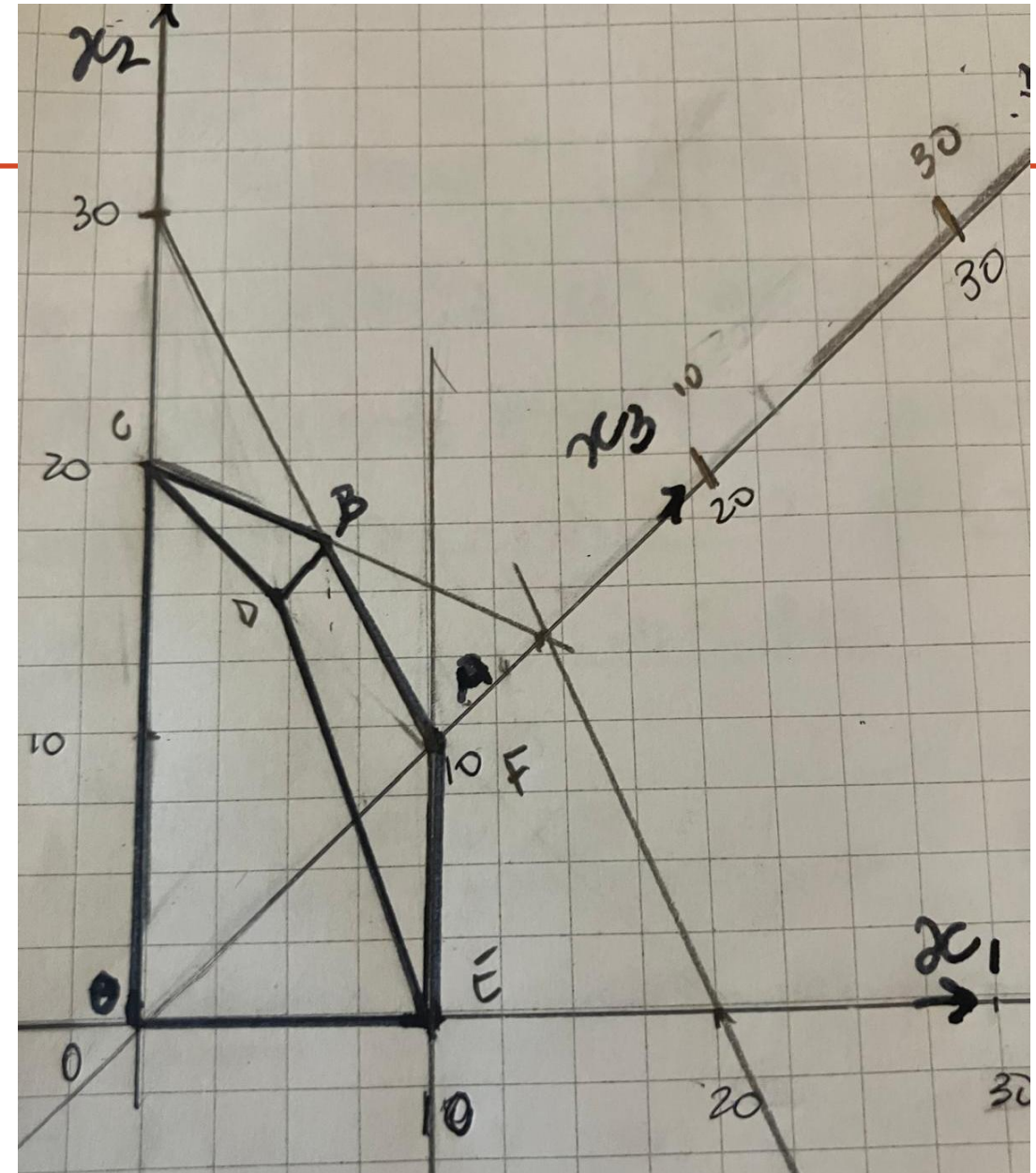
$$x \geq 0 \text{ where } c = [c_1, c_2, \dots, c_n],$$

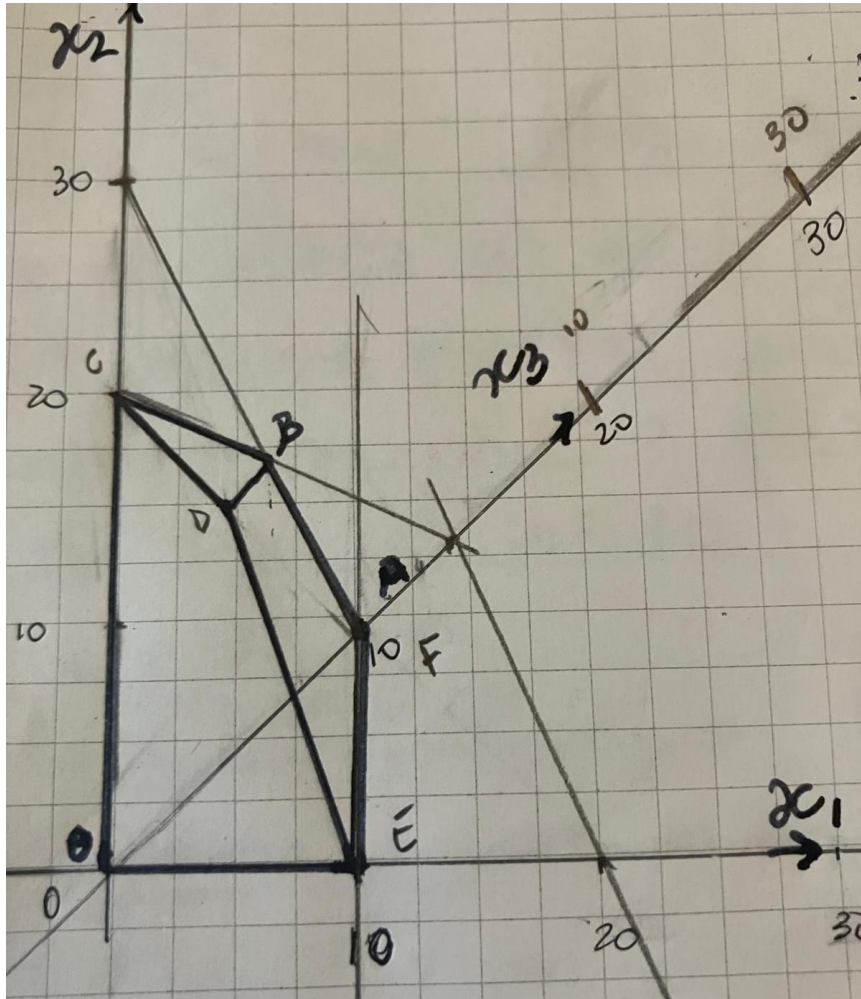
$$\begin{bmatrix} Z \\ x_B \end{bmatrix} = \begin{bmatrix} 1 & c_B B^{-1} \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ b \end{bmatrix} = \begin{bmatrix} c_B B^{-1} b \\ B^{-1} b \end{bmatrix}$$

$$\begin{bmatrix} 1 & c_B B^{-1} \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} 1 & -c & 0 \\ 0 & A & I \end{bmatrix} = \begin{bmatrix} 1 & c_B B^{-1} A - c & c_B B^{-1} \\ 0 & B^{-1} A & B^{-1} \end{bmatrix}$$



$$\begin{bmatrix} 1 & -c & 0 \\ 0 & A & I \end{bmatrix} \begin{bmatrix} Z \\ x \\ x_s \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix}.$$





In-class exercise

From book

D 5.3-1.* Consider the following problem.

$$\text{Maximize } Z = x_1 - x_2 + 2x_3,$$

subject to

$$2x_1 - 2x_2 + 3x_3 \leq 5$$

$$x_1 + x_2 - x_3 \leq 3$$

$$x_1 - x_2 + x_3 \leq 2$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

Let x_4 , x_5 , and x_6 denote the slack variables for the respective constraints. After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of:							Right Side
		Z	x_1	x_2	x_3	x_4	x_5	x_6	
Z	(0)	1				1	1	0	
x_2	(1)	0				1	3	0	
x_6	(2)	0				0	1	1	
x_3	(3)	0				1	2	0	

- Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.
- Identify the defining equations of the CPF solution corresponding to the optimal BF solution in the final simplex tableau.

Read Chapter 6.1 – 6.3; 8.1

Duality Theory

Steps to Transform a LP in standard form to its dual:

Associate every functional constraint i with a dual variable y_i

Construct the minimizing (dual) objective function y_0 as the sum of the product of every RHS b_i multiplied by the associated variable y_i .

For every initial variable x_j , construct a dual constraint by multiplying each y_i by the coefficient of x_j in the i th constraint, a_{ij} , summing up all terms and setting the sum to " $\geq$ " the coefficient of x_j in the objective function, c_j .

Set all y_i 's to " ≥ 0 "

Dual of the Dual: Prototype Example

(P)

$$\begin{aligned} \text{Max } z &= 3x_1 + 5x_2 \\ \text{s.t. } x_1 &\leq 4 \\ &2x_2 \leq 12 \\ 3x_1 + 2x_2 &\leq 18 \\ x_1, x_2 &\geq 0 \end{aligned}$$



Dual (P)

$$\begin{aligned} \text{Min } y_0 &= 4y_1 + 12y_2 + 18y_3 \\ \text{s.t. } y_1 &+ 3y_3 \geq 3 \\ &2y_2 + 2y_3 \geq 5 \\ y_1, y_2, y_3 &\geq 0 \end{aligned}$$



$$\begin{aligned} \text{Max } -w &= 3u_1 + 5u_2 \\ \text{s.t. } u_1 &\leq 4 \\ &2u_2 \leq 12 \\ 3u_1 + 2u_2 &\leq 18 \\ u_1, u_2 &\geq 0 \end{aligned}$$

Dual [Dual(P)]

$$\begin{aligned} \text{Min } w &= -3u_1 - 5u_2 \\ \text{s.t. } -u_1 &\geq -4 \\ &-2u_2 \geq -12 \\ -3u_1 - 2u_2 &\geq -18 \\ u_1, u_2 &\geq 0 \end{aligned}$$

Equivalent to Dual(P) in standard form

$$\begin{aligned} \leftarrow \text{Max } -y_0 &= -4y_1 - 12y_2 - 18y_3 \\ \text{s.t. } -y_1 &- 3y_3 \leq -3 \\ &-2y_2 - 2y_3 \leq -5 \\ y_1, y_2, y_3 &\geq 0 \end{aligned}$$

Dual of the Dual: General Problem

